CAPSTONE PROJECT REPORT

on

Bash Scripting Suite for System Maintenance

Submitted by

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1. Project Objective

To design and implement a lightweight and efficient Bash scripting toolkit that automates essential system maintenance tasks — including system monitoring, disk analysis, and process management — to enhance system reliability, reduce manual effort, and demonstrate automation concepts using Linux shell scripting.

2. Introduction

The Bash Scripting Suite for System Maintenance is developed to simplify routine system administration tasks on Linux.
It integrates multiple modular shell scripts into a single interactive framework (main.sh) that automates maintenance operations such as:

- Automated backup creation
- System update and cleanup
- Log monitoring
- Disk usage analysis
- System information reporting

This project demonstrates Linux automation through Bash scripting, emphasizing file handling, process monitoring, conditional logic, and modular design.

3. Daily Task Report

Day Task Description Outcome

Successfully created the workspace
Day Project setup, folder structure creation, and Git
BashMaintenanceSuite and
initialized 1 repository initialization.
Git.

Developed backup.sh and update-clean.sh scripts
Day Automated backup and update modules for automated backup and system update
2 completed and tested. operations.

Implemented logwatch.sh and integrated logging
Day to monitor system logs and record activity in the Log monitoring automation
achieved.
3 logs/ directory.

Added new modules: diskinfo.sh and sysinfo.sh
Day Complete maintenance suite with for disk and system information reporting;
4 disk/system info automation built. integrated all into main.sh menu.

Day Tested all scripts, improved error handling, and Final version tested successfully
and 5 finalized the interactive menu-based execution. uploaded to GitHub.

4. Methodology

Planning & Setup

- Defined project objectives and scope.
- Created folder structure: scripts/, logs/, and main menu script.
- Initialized Git repository for version control.

Script Development

- Developed and tested individual Bash scripts for: ○ Backup (backup.sh) ○
  Update & cleanup (update-clean.sh) ○ Log monitoring (logwatch.sh) ○
  Disk usage (diskinfo.sh) ○ System information (sysinfo.sh)

Integration

- Integrated all scripts into main.sh using a menu-driven interface with error handling and logging.

5. Results

- Full automation achieved using modular Bash script.
- Dynamic logs generated for every operation in logs/.
- Backups automatically stored under ~/backups/ with timestamps.
- Gained hands-on understanding of Linux commands such as rsync, apt, df, lscpu, and process monitoring utilities.
- • Improved understanding of Linux process, disk, and system monitoring commands.

Code Sections

Backup.sh

Automated Backup Script — Uses rsync to back up user data to a timestamped directory and logs the output.

```
#!/bin/bash
# Automated Backup Script
# Author: Pradyush Mohapatra

BLUE=$(tput setaf 6)
GREEN=$(tput setaf 2)
RESET=$(tput sgr0)

echo "${BLUE}=============================================="
echo "        Automated System Backup"
echo "==============================================${RESET}"

SOURCE="$HOME/Documents"
DEST="$HOME/backups/backup_$(date +%Y-%m-%d_%H-%M-%S)"
LOG_DIR="$(dirname "$0")/../logs" mkdir
-p "$DEST" "$LOG_DIR"

echo "Starting backup from $SOURCE to $DEST ..."
rsync -avh --progress "$SOURCE/" "$DEST/" &>> "$LOG_DIR/backup.log"
 echo "${GREEN} Backup completed successfully!"
echo "Files saved in: $DEST"
echo "Log saved at: $LOG_DIR/backup.log${RESET}"
```

update-clean.sh

system Update & Cleanup Script — Performs package update,

upgrade, and cleanup using APT

```
#!/bin/bash
# System Update and Cleanup Script
# Author: Pradyush Mohapatra

BLUE=$(tput setaf 6)
GREEN=$(tput setaf 2)
RESET=$(tput sgr0)
```

```
echo "${BLUE}=============================================="
echo "        System Update and Cleanup"
echo "==============================================${RESET}"

LOG_DIR="$(dirname "$0")/../logs"
mkdir -p "$LOG_DIR"
 {   sudo apt update -y
sudo apt upgrade -y   sudo
apt autoremove -y   sudo
apt autoclean -y } &>>
"$LOG_DIR/update.log"
 echo "${GREEN} System updated and cleaned
successfully!${RESET}" echo "Log saved at: $LOG_DIR/update.log"
```

disk_info.s
h

Disk Usage Monitoring Script: Displays disk size, used, available space, and percent used.

```
#!/bin/bash
# Disk Storage Information Script
# Author: Pradyush Mohapatra

BLUE=$(tput setaf 6)
GREEN=$(tput setaf 2)
RESET=$(tput sgr0)

echo "${BLUE}=============================================="
echo "        Disk Storage Information"
echo "==============================================${RESET}"

LOG_DIR="$(dirname "$0")/../logs"
mkdir -p "$LOG_DIR"

df -h --output=source,size,used,avail,pcent | grep -vE 'tmpfs|udev' | tee
"$LOG_DIR/diskinfo.log"
 echo "${GREEN} Disk information fetched
successfully!${RESET}" echo "Log saved at:
$LOG_DIR/diskinfo.log"
```

logwatch.sh

Log Monitoring Script — Scans /var/log/syslog for errors, warnings, or failures.

```
#!/bin/bash
# Log Monitoring Script
# Author: Pradyush Mohapatra

BLUE=$(tput setaf 6)
GREEN=$(tput setaf 2)
RESET=$(tput sgr0)

echo "${BLUE}=============================================="
echo "        Log Monitoring and Alerts"
echo "==============================================${RESET}"

LOG_DIR="$(dirname "$0")/../logs"
mkdir -p "$LOG_DIR"

grep -Ei "error|fail|warning" /var/log/syslog > "$LOG_DIR/logwatch.log" || true
 echo "${GREEN} Log scan completed!${RESET}"
echo "Results stored in:
$LOG_DIR/logwatch.log"
```

sysinfo.sh

System Information Script — Displays hostname, OS, kernel, architecture, uptime, and memory usage.

```
#!/bin/bash
# System Information Script
# Author: Pradyush Mohapatra

BLUE=$(tput setaf 6)
GREEN=$(tput setaf 2)
RESET=$(tput sgr0)

echo "${BLUE}=============================================="
echo "        System Information"
echo "==============================================${RESET}"

LOG_DIR="$(dirname "$0")/../logs"
mkdir -p "$LOG_DIR"
 {
  echo "Hostname: $(hostname)"
  echo "Operating System: $(lsb_release -d 2>/dev/null | cut -
f2)"   echo "Kernel: $(uname -r)"   echo "Architecture: $(uname
-m)"   echo "Uptime: $(uptime -p)"   echo "Logged-in Users:
$(who | wc -l)"   echo "Memory Usage:"   free -h
} | tee "$LOG_DIR/sysinfo.log"
 echo "${GREEN} System information fetched
successfully!${RESET}" echo "Log saved at: $LOG_DIR/sysinfo.log"
```

main.s
h

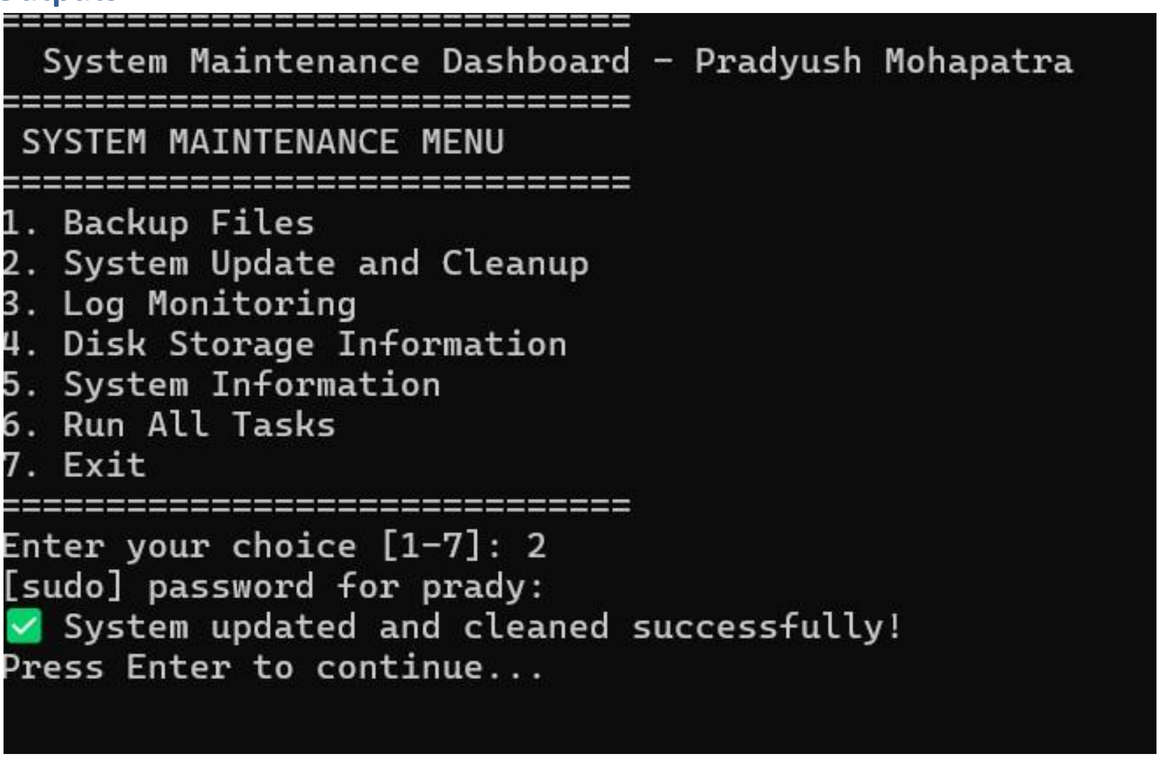
```
#!/usr/bin/env bash
# ==========================================
# Script Name : main.sh
# Author     : <Your Name> (<Your Roll No>)
# Description : Interactive menu for maintenance suite
# ==========================================

set -euo pipefail

BASE_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$BASE_DIR/scripts
" LOG_DIR="$BASE_DIR/logs"
mkdir -p "$LOG_DIR"
 while true; do   clear   echo
"========================================================"
echo "  System Maintenance Dashboard - Pradyush Mohapatra"
echo "========================================================"
echo " SYSTEM MAINTENANCE MENU"   echo
"=============================="   echo "1. Backup Files"
echo "2. System Update and Cleanup"
 echo "3. Log Monitoring"   echo "4. Disk
Storage Information"   echo "5. System
Information"   echo "6. Run All Tasks"
echo "7. Exit"   echo
"=============================="   read -rp
"Enter your choice [1-7]: " choice
```

```
 case "$choice" in    1)      bash
"$SCRIPTS_DIR/backup.sh"      read -rp
"Press Enter to continue..."
    ;;    2)      bash
"$SCRIPTS_DIR/update-clean.sh"      read
-rp "Press Enter to continue..."
    ;;    3)      bash
"$SCRIPTS_DIR/logwatch.sh"      read -rp
"Press Enter to continue..."
    ;;    4)      bash
"$SCRIPTS_DIR/diskinfo.sh"      read -rp
"Press Enter to continue..."
    ;;    5)      bash
"$SCRIPTS_DIR/sysinfo.sh"      read -rp
"Press Enter to continue..."
    ;;    6)      bash
"$SCRIPTS_DIR/backup.sh"      bash
"$SCRIPTS_DIR/update-clean.sh"      bash
"$SCRIPTS_DIR/logwatch.sh"      bash
"$SCRIPTS_DIR/diskinfo.sh"      bash
"$SCRIPTS_DIR/sysinfo.sh"      echo "  All
maintenance tasks completed!"      read -rp
"Press Enter to continue..."
    ;;    7)      echo "Exiting...
Goodbye!"      exit 0      ;;    *)
echo "Invalid choice. Try again."
sleep 1      ;;   esac done
```

Outputs



```
================================
  System Maintenance Dashboard - Pradyush Mohapatra
================================
 SYSTEM MAINTENANCE MENU
================================
```

```
itesh@Ritesh:~/BashScriptingSuite$ ./system_info.sh
=============================================
          System Information
=============================================
Hostname       : Ritesh
OS Version     : Ubuntu 24.04.1 LTS
Kernel Version : 4.4.0-19041-Microsoft
Architecture   : x86_64
Uptime         : up 11 minutes
CPU Model      : Intel(R) Core(TM) i7-5600U CPU @ 2.60GHz
Total Memory   : 3.9Gi
Used Memory    : 3.2Gi
Free Memory    : 685Mi
Logged-in Users: 0
Current User   : ritesh
---------------------------------------------
 System information fetched successfully!
```

```
================================
  System Maintenance Dashboard - Pradyush Mohapatra
================================
 SYSTEM MAINTENANCE MENU
================================
1. Backup Files
2. System Update and Cleanup
3. Log Monitoring
4. Disk Storage Information
5. System Information
6. Run All Tasks
7. Exit
================================
Enter your choice [1-7]: 3
✅ Log monitoring completed. See logs/logwatch.log
Press Enter to continue...
```

```
==============================
 SYSTEM MAINTENANCE MENU
==============================
. Backup Files
. System Update and Cleanup
. Log Monitoring
. Disk Storage Information
. System Information
. Run All Tasks
. Exit
==============================
nter your choice [1-7]: 4
✅ Disk Storage Information:
ilesystem      Size  Used Avail Use%
one            3.8G     0  3.8G   0%
one            3.8G  4.0K  3.8G   1%
rivers         232G  217G   15G  94%
dev/sdd       1007G  1.7G  955G   1%
one            3.8G   84K  3.8G   1%
one            3.8G     0  3.8G   0%
ootfs          3.8G  2.7M  3.8G   1%
one            3.8G  548K  3.8G   1%
one            3.8G     0  3.8G   0%
one            3.8G     0  3.8G   0%
one            3.8G   76K  3.8G   1%
one            3.8G   76K  3.8G   1%
:\             232G  217G   15G  94%
:\             244G   54G  190G  23%
ogs saved to /home/prady/BashMaintenanceSuite/scripts/../logs/diskinfo.log
ress Enter to continue...
```

```
==============================
 System Maintenance Dashboard - Pradyush Mohapatra
==============================
SYSTEM MAINTENANCE MENU
==============================
. Backup Files
. System Update and Cleanup
. Log Monitoring
. Disk Storage Information
. System Information
. Run All Tasks
. Exit
==============================
nter your choice [1-7]: 5
✅ System Information:
perating System: Ubuntu 24.04.3 LTS
         Kernel: Linux 6.6.87.2-microsoft-standard-WSL2
   Architecture: x86-64
ogs saved to /home/prady/BashMaintenanceSuite/scripts/../logs/sysinfo.log
ress Enter to continue...
```